University of Florida, Department of Mathematics

Numerical Analysis PhD Exam, January 2016, Part 2

You must show all your work as neatly and clearly as possible and indicate the final answer clearly.

1. A fourth degree polynomial $P(x)$ satisfies $\Delta^4 P(0) = 24$, $\Delta^3 P(0) = 6$, and $\Delta^2 P(0) = 0$, where $\Delta P(x) = P(x+1) - P(x)$. Compute $\Delta^2 P(10)$.

2. Consider the fixed point iteration

$$x_{n+1} = \phi(x_n), \qquad n = 0, 1, 2, ...$$

where

$$\phi(x) = Ax + Bx^2 + Cx^3.$$

Given a positive number α, determine the constants A, B, C such that the iteration converges locally to $1/\alpha$ with order $p = 3$.

3. This problem has the following parts:

(a) Find α, β and γ so that the quadrature formula has a maximum degree of precision. What is the exact degree of precision of this quadrature formula?

$$\int_0^2 f(x)\,dx \approx \alpha(f(0) + f(2)) + \beta(f'(0) - f'(2)) + \gamma(f''(0) - f''(2)).$$

(b) Suggest your own quadrature formula for the integral

$$\int_0^2 f(x)\,dx.$$

Your formula should have expected degree of precision at least four (you don't have to compute to demonstrate that).

4. For the function

$$f(x) = \sqrt[m]{|x|}, \qquad m \neq \frac{1}{2},$$

in the interval $[-1, 1]$, find the polynomial of minimax approximation of degree 2. What is the minimax error? What happens if $m = \frac{1}{2}$?

5. Prove that Gaussian quadrature formula (for any n) in the interval $[-1, 1]$ and with weight $w(x) = 1$ is exact for all odd functions.